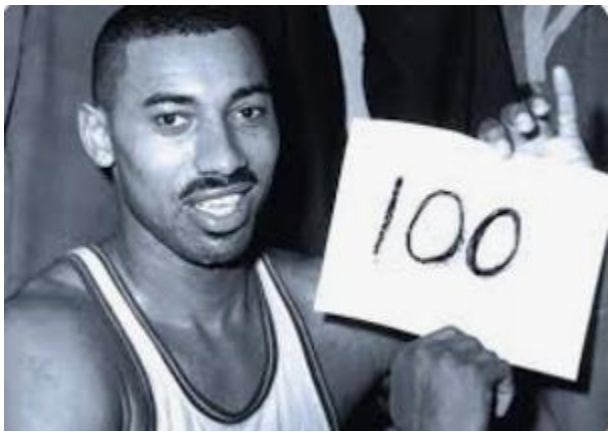

STAT 410 PROJECT

Modeling Basketball Production

(Zero-Inflated Poisson Process)

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Introduction

The NBA is a player-driven league that is both fun to watch and produces a ton of performance-based statistics in every game. The fun part for statisticians is figuring out which statistics are "empty calories" and which contribute to positive team performance. Advanced stats like PER (player efficiency rating) and WAR (wins above replacement) are the first steps in quantifying a player's "worth". By modeling points scored in a game by individual players with a specialized Poisson distribution, we can better understand how common each performance is and see which explanatory variables significantly affect the model.

Zero-Inflated Poisson Background

15 players (maximum) on each team suit up for each NBA game and not all of them score points, but this can be for a myriad of reasons. We theorize that individual points scored can best be modeled by a Zero-Inflated Poisson Process: there are structural zeros of points scored that come from players not playing significant minutes (from injury or strategic team matchups) but also chance zeros, that come from players who do play non-zero minutes but perform poorly/miss all their shots. Additionally, points scored in a game is also affected by other variables, like player position, height, and years of basketball experience. Therefore, a Zero-Inflated Poisson model of the distribution of points scored will best explain our outcomes.

Data Description

To model points scored in NBA games, we found an enormous data set on Kaggle that contained all NBA games played from 2004 to winter 2022. To limit the scope of this project and use the most recent data possible, we restricted our observations to the 200 most recent player-performance observations, which consisted of 8 regular season games between 16 rosters on December 21st and 22nd, 2022. The player position variable has some missingness that we had to manually fill, and we added data for player height and years played. The variables of interest for our 200 observations are Points Scored, Starting Position/Position Played (character value of C, F, or G), Minutes Played, Plus/Minus, Height, and Years in the League (as of 2022).

Results

The Zero-Inflated Poisson model proved to be a very good model due to the very low p-value (< .001) on the deviance test in both R and SAS. Additionally, the histogram of points scored had a very clear ZIP model "shape" (See Figure 1) with a pronounced column of zero values containing a third of the data and then a more typical poisson distribution in the non-zero columns.

We proceed with the Zero-Inflated Poisson, the fitted model for the data is:

π-hat = (e ^ (2.99186 - .42872 * *MinutesPlayed*)) /

(1 + e ^ (2.99186 - .42872 * *MinutesPlayed*))

and

λ-hat = e ^ (1.491997 + .080807 * *Forward* + .152739 * *Guard* + .132361 * *Height* + .012162 * *PlusMinus* + .003644* *YearsPlayed*)

The interpretations for the significant predictors at a 5% level are as follows: As minutes played increases by one minute, the estimated odds of scoring zero points change by (e^(.-.42872) -1) * 100% = - 34.8657% , which is to say decrease by 34.8657%. PlusMinus is our only variable that has a significant effect on the average points scored in a game. When PlusMinus increases by 1 point, the expected points scored in the game go up by (e^(.012162) -1) * 100% = 1.2235 %.

At the 5% significant level Minutes Played (p < .001) significantly predicts the odds of scoring zero points. This is excellent because intuitively it must have a strong correlation: players that do not play minutes cannot score. Even though this relationship is clear, using minutes to model structural zeros adds significant power to our model's predictive capabilities. We see that minutes played increasing by just one causes the odds of scoring zero points to plummet by 34.87 %.

Additionally, PlusMinus has a significant effect on the points scored in a game, while Position, Height, and Years in the League did not have a significant effect, all of which are interesting results. PlusMinus has a slight effect (a point increase in plus minus leads to a 1.22% increase in expected points scored) but is an extremely significant predictor (p < .001). PlusMinus is a measure of how much better your team performed than the other team while the

player was on the court, so it's interesting that it has such a strong correlation to an individual stat like Points that is simply raw output and has nothing to do with relative performance. Conversely, finding that position played, height, and years in the league were NOT significant predictors of points scored was also surprising, I could see an argument for all three of these variables contributing to scoring load. While a variable like height is clearly valuable for basketball, perhaps players that have made it to the NBA are effective across the board which ends up controlling for these variables. To say this another way, the bias for height in basketball has already been corrected for in the NBA player selection process, so within the game there is no skew towards taller players performing better.

When using the model for prediction, I selected a random player performance from the greater Kaggle list using a random number generated with R. I predicted player performance 731 by T.J. Warren: Forward, 6.66 feet tall, 9th year in the league, PlusMinus of -5, played 9.25 minutes. Plugging this into our model we got an expected scored of 8.2148 points in the game, which was fairly close to our actual value of 6 points scored.

Conclusion

The nba dataset and Zero-Inflated Poisson model did a satisfactory job of modeling the points scored distribution very closely. We were able to discern which variables had very significant predictive power (Minutes and PlusMinus) and which had almost no predictive power (Height, Position, YearsLeague). Minutes played had a very large effect on the expected probability of scoring zero points, and PlusMinus had a moderate/minimal positive effect on the expected value of points scored. I was also able to predict the probability of a random selected player performance quite accurately.

Appendix-

A. Zero-Inflated Poisson SAS Code

```
proc import datafile="//vdi-fileshare02/UEMprofiles/030870968/Desktop/410nbasas.csv"
      out=nbadata
      dbms=csv
      replace;
      getnames=yes;
  run;
proc print data=nbadata; /* To view the imported data */
  run;
proc genmod;
  class Position(ref="C");
  model Pts=Position PlusMinus Height YearsLeague/dist=zip;
  zeromodel Min;
  run;
proc genmod;
  model Pts=/dist=zip;
  zeromodel;
  run;
data deviance_test;
  deviance=-2*(-71.3892-(-57.0406));
  pvalue=1-probchi(deviance,3);
  run;
```

B. SAS Output

Analysis Of Maximum Likelihood Parameter Estimates								
Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	1.4920	0.8533	-0.1804	3.1643	3.06	0.0804
Position	F	1	0.0808	0.0745	-0.0653	0.2269	1.18	0.2783
Position	G	1	0.1527	0.0986	-0.0405	0.3460	2.40	0.1214
Position	C	0	0.0000	0.0000	0.0000	0.0000	.	.
PlusMinus		1	0.0122	0.0020	0.0082	0.0161	36.38	<.0001
Height		1	0.1324	0.1233	-0.1093	0.3740	1.15	0.2831
YearsLeague		1	0.0036	0.0060	-0.0081	0.0153	0.37	0.5416
Scale		0	1.0000	0.0000	1.0000	1.0000		

Analysis Of Maximum Likelihood Zero Inflation Parameter Estimates							
Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept	1	2.9919	0.5802	1.8546	4.1291	26.59	<.0001
Min	1	-0.4287	0.0765	-0.5787	-0.2787	31.37	<.0001

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance		1733.4461	
Scaled Deviance		1733.4461	
Pearson Chi-Square	198	493.3694	2.4918
Scaled Pearson X2	198	493.3694	2.4918
Log Likelihood		2617.6649	

elihood Parameter Estimates			
% Confidence Limits		Wald Chi-Square	Pr > ChiSq
.4517	2.5435	11371.1	<.0001
.0000	1.0000		

d Zero Inflation Parameter Estimates			
% Confidence Limits		Wald Chi-Square	Pr > ChiSq
.4187	-0.7786	45.26	<.0001

C. Zero-Inflated Poisson R Code

```
nba.data<- read.csv(file="C:/Users/jknau/OneDrive/Desktop/410 Applied Regression/410nba.csv", header=TRUE, sep=",")
summary(nba.data)
```

```
hist(nba.data$Pts,
     breaks = seq(0, 60 , by = 2),
     main = "Histogram of Points Scored",
     xlab = "Pts",
     ylab = "Frequency",
     col = "salmon2",
     border = "black")
```

```
nba.data$Position <- factor(nba.data$Position)
#specifying reference category
nba.rel<- relevel(nba.data$Position, ref="G")
library(pscl)
summary(fitted.model<- zeroinfl(Pts ~ Position + Height + PlusMinus + YearsLeague|Min,
                                data=nba.data))
null.model<- zeroinfl(Pts ~ 1, data=nba.data)
print(deviance<- -2*(logLik(null.model)-logLik(fitted.model)))
print(p.value<- pchisq(deviance, df=3, lower.tail=FALSE))
```

```
> print(deviance<- -2*(logLik(null.model)-logLik(fitted.model)))
'log Lik.' 213.1957 (df=2)
> print(p.value<- pchisq(deviance, df=3, lower.tail=FALSE))
'log Lik.' 5.935921e-46 (df=2)
```

```
> random_number <- sample(1:1000, 1)
> print(random_number)
[1] 731
```

D. R Output

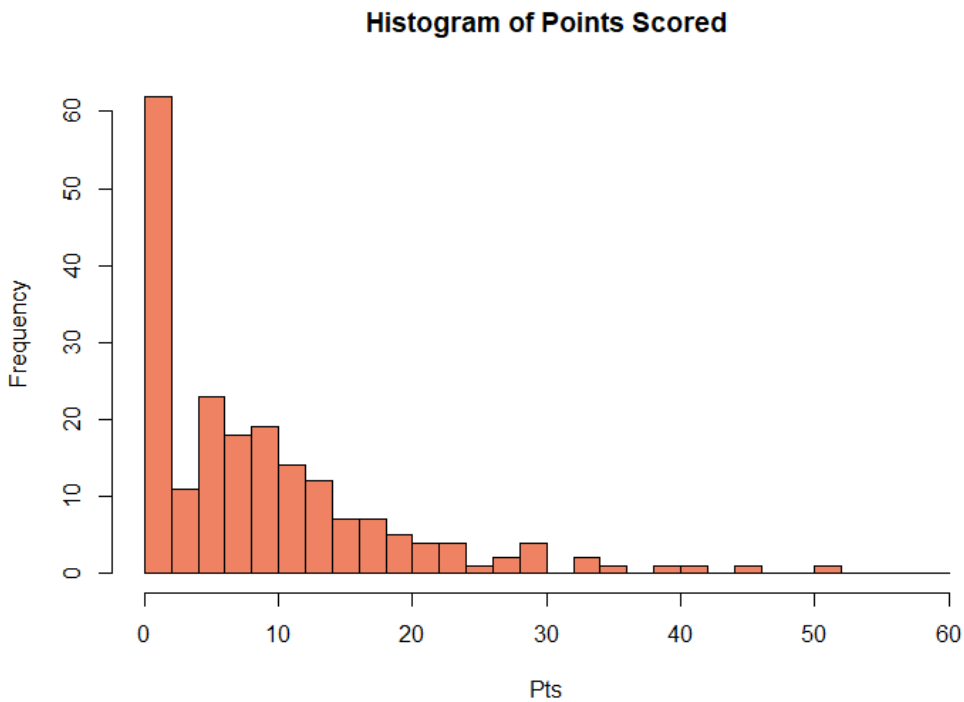

Figure 1: Points Histogram

```
   PlusMinus     YearsLeague       Pts              Min              Height
 Min.   :-31   Min.   : 1.00   Min.   : 0.000   Min.   : 0.000   Min.   :5.920
 1st Qu.: -6   1st Qu.: 2.00   1st Qu.: 1.500   1st Qu.: 5.518   1st Qu.:6.383
 Median :  0   Median : 5.00   Median : 7.000   Median :21.090   Median :6.670
 Mean   :  0   Mean   : 5.66   Mean   : 9.115   Mean   :18.859   Mean   :6.579
 3rd Qu.:  5   3rd Qu.: 7.00   3rd Qu.:13.000   3rd Qu.:29.030   3rd Qu.:6.830
 Max.   : 31   Max.   :16.00   Max.   :52.000   Max.   :45.370   Max.   :7.080
```

```
Pearson residuals:
    Min      1Q  Median      3Q     Max
-3.3505 -1.1712 -0.2851  0.4741 10.5056

Count model coefficients (poisson with log link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 1.491997   0.853254   1.749   0.0804 .
PositionF   0.080807   0.074528   1.084   0.2783
PositionG   0.152739   0.098612   1.549   0.1214
Height      0.132361   0.123306   1.073   0.2831
PlusMinus   0.012162   0.002016   6.031 1.62e-09 ***
YearsLeague 0.003644   0.005970   0.610   0.5416

Zero-inflation model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.99186    0.58024   5.156 2.52e-07 ***
Min         -0.42872    0.07654  -5.601 2.13e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 17
Log-likelihood: -760.1 on 8 Df
```

```
> print(predict(fitted.model, data.frame(Position="F", Height=6.66,YearsLeague=9,
+                                        PlusMinus=-5, Min= 9.25)))
       1
8.214772
```

E. SAS CODE PREDICTION

```
data prediction;
input Position$ Min PlusMinus Height YearsLeague;
cards;
F 9.25 -5 6.66 9
;
data nbadata;
set nbadata prediction;
run;
proc genmod;
class Position(ref="C");
model Pts=Position PlusMinus Height YearsLeague/dist=zip;
zeromodel Min;
output out=outdata p=pred_pts;
run;
proc print data=outdata(firstobs=202)noobs;
var pred_pts;
run;
```

F. SAS CODE PREDICTION OUTPUT

The SAS System

pred_pts
8.21477

G. R CODE PREDICTION

```
> print(predict(fitted.model, data.frame(Position="F", Height=6.66,YearsLeague=9,
+                                        PlusMinus=-5, Min= 9.25)))
```

H. R CODE PREDICTION OUTPUT

```
+
       1
8.214772
```

References

Nathan Lauga (2004-2020). (2020, December). *NBA games data*. Kaggle. Retrieved December 6th, 2024. https://www.kaggle.com/datasets/nathanlauga/nba-games/data